

- 14** A red indicator solution was mixed with agar, and the resulting solid was cut into small cuboid blocks. The blocks were placed in an acid which turns the indicator yellow, and all other variables were kept constant. The dimensions of the three blocks used are shown.

block 1 $3\text{ mm} \times 3\text{ mm} \times 3\text{ mm}$

block 2 $8\text{ mm} \times 8\text{ mm} \times 8\text{ mm}$

block 3 $11\text{ mm} \times 11\text{ mm} \times 11\text{ mm}$

Which row shows the correct surface area (SA) to volume (V) ratio for each block, and the time taken for the block to turn yellow?

	block 1		block 2		block 3	
	SA to V ratio	time to turn yellow / mins	SA to V ratio	time to turn yellow / mins	SA to V ratio	time to turn yellow / mins
A	0.5 : 1.0	4	1.33 : 1.0	11	1.83 : 1.0	13
B	0.5 : 1.0	13	1.33 : 1.0	11	1.83 : 1.0	4
C	2.0 : 1.0	4	0.75 : 1.0	11	0.55 : 1.0	13
D	2.0 : 1.0	13	0.75 : 1.0	11	0.55 : 1.0	4